



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	CSC3001
Course title (English)	Discrete Mathematics
Course title (Chinese)	离散数学
Units	3
Language of Instruction	English
Description (English)	This course introduces the relevant mathematical concepts and techniques in computer science with emphasis on proofs and rigorous reasoning. Topics include: logics, methods of proofs, recursion, elementary number theory, graph theory, counting.
Description (Chinese)	本课程介绍与计算机科学密切相关的，离散而非连续的数学概念与方法，强调证明及逻辑推理。内容包括：逻辑，证明方法，递归，初等数论，图论，计数方法等。

2. Prerequisites / Co-requisites

A. Prerequisites

No prerequisites, however this course would presume some basics of MAT2040.

B. Co-requisites

3. Learning Outcomes

Knowledge:

- Students will understand the basic concepts and operators in logics.
- Students will learn the basic methods of proofs.
- Students will learn the classical results in recursions.
- Students will learn some fundamental result on the greatest common divisors, the concept of modular arithmetic, and Chinese remainder theorem.
- Students will understand some basic concepts in graph theory and some of the



fundamental results.

f) Students will learn some commonly used counting methods.

Skills

Generic:

g) Students will be able to translate daily life language into a logical formula and simplify it.

h) Students will be able to apply various proof methods to complete a rigorous proof.

i) Students will be able to apply the fundamental knowledge of number theory to solve some real-life problems.

j) Students will be able to model some real-life problems as a graph problem.

Valued/Attitude:

k) Students will have a clear mind to tackle complicated problems.

l) Students will be able to analyze many real-life problems from the viewpoint of mathematics.

m) Students will have awareness of the impact of mathematics in computer science and other related areas.

n) Students will be more literate in science. They will develop a knowledge in science so that they can disseminate science (or pseudo-science) related news articles.

4. Course syllabus

Propositional Logic: mathematical proof, logic, basic operators, using simple operators to construct any operator, logic equivalence, De Morgan's law, conditional statement, arguments

Sets, First-Order Logic: definitions, operations on sets, set identities, Russell's paradox, quantifiers, negation, multiple quantifiers, arguments of quantified statements

Methods of Proofs: direct proof, contrapositive, proof by contradiction, proof by cases

Mathematical Induction: the idea of induction, basic induction proofs, inductive constructions, paradox, strong induction, well ordering principle, invariant method

Recursion: Fibonacci recurrence, problem solving recurrences, Catalan number recurrences, tower of Hanoi, merge sort, second order recurrence relation, generating functions, distinct-roots theorem

Greatest Common Divisors: common divisors, the quotient-remainder theorem, Euclidean GCD algorithm, prime divisibility, fundamental theorem of arithmetic

Modular Arithmetic, Chinese Remainder theorem: modular addition, modular multiplication, multiplicative inverse, cancellation, Fermat's little theorem, Wilson's theorem

Graphs: seven bridges of Königsberg, graphs, degrees, isomorphism, path, cycle, connectedness, trees, Eulerian cycle, directed graphs

Graph Matching: stable matching problem and Hall's theorem, reduction and applications, proof of Hall's theorem

Graph Coloring: graph coloring, applications, some positive results, planar graphs, Euler's formula, 6-coloring



Combinatorial Proofs and its Principles: binomial coefficients, combinatorial proof, inclusion-exclusion formula, pigeonhole principle and applications

Counting by Mapping: definition of functions, injection, surjection, bijection, inverse function, composite, bijection rule, division rule, Catalan number

Cardinality (optional): cardinality, countable, uncountable, diagonal argument.

5. Assessment Scheme

Component/ method	% weight
Assignments	20
Mid-Term Exam	30
Final Exam	50

6. Grade descriptor

General

Grade	Description
A	Demonstrates the ability to achieve all the learning outcomes in this course and be able to apply the principles in solving unfamiliar problems in novel situations, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet



specified assessment requirements.

7. Feedback for evaluation

- Communicate with instructor and/or TAs by chatting, emails etc.
- Evaluate student performance through assignments, mid-term and final exams
- Collect student feedback from CTE by the end of the course
- Collect opinions from peers by internal and external course reviews

8. Reading

9. Course components

Activity	Hours/week
Lecture	3
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
001	Propositional Logic: mathematical proof, logic, basic operators, using simple operators to construct any operator, logic equivalence, De Morgan's law, conditional statement, arguments
002	Sets, First Order Logic: definitions, operations on sets, set identities, Russell's paradox, quantifiers, negation, multiple quantifiers, arguments of quantified statements
003	Methods of Proofs: direct proof, contrapositive, proof by contradiction, proof by cases
004	Mathematical Induction: the idea of induction, basic induction proofs, inductive constructions, paradox, strong induction, well ordering principle, invariant method
005	Recursion: Fibonacci recurrence, problem solving recurrences, Catalan number recurrences, tower of Hanoi, merge sort, second order recurrence relation, generating functions, distinct-roots theorem
006	Greatest Common Divisors: common divisors, the quotient-remainder theorem, Euclidean GCD algorithm, prime divisibility, fundamental theorem of arithmetic
007	Modular Arithmetic, Chinese Remainder theorem: modular addition, modular multiplication, multiplicative inverse, cancellation, Fermat's little theorem, Wilson's theorem



008	Mid-Term, other topics in number theory (optional)
009	Graphs: seven bridges of Königsberg, graphs, degrees, isomorphism, path, cycle, connectedness, trees, Eulerian cycle, directed graphs
010	Graph Matching: stable matching problem and Hall's theorem, reduction and applications, proof of Hall's theorem
011	Graph Coloring: graph coloring, applications, some positive results, planar graphs, Euler's formula, 6-coloring
012	Combinatorial Proofs and its Principles: binomial coefficients, combinatorial proof, inclusion-exclusion formula, pigeonhole principle and applications
013	Counting by Mapping: definition of functions, injection, surjection, bijection, inverse function, composite, bijection rule, division rule, Catalan number
014	Cardinality (optional): cardinality, countable, uncountable, diagonal argument. Revision

11. Any other information

12. Version date

Version number	5
As of (date)	2024/06/07